Announcements

Final exam: Tuesday, 5/12, 8:00am–11:00am, Everitt 3117
Covers entire course
Policy email to come

Wednesday's class: review (no quiz)

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§11.2: Applications of trees

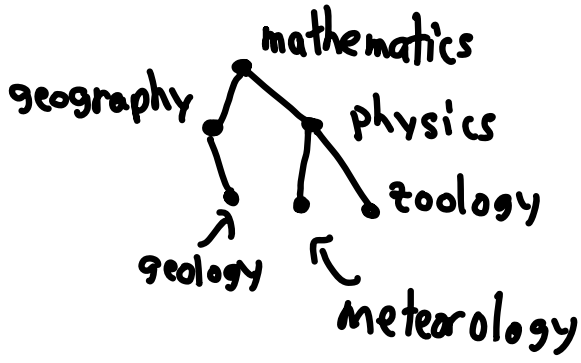
Binary search trees

Suppose we have a list of words, which we want in alphabetical order

Add them to a binary tree, such that

left child < parent < right child

Ex 1: {mathematics, physics, geography, zoology, meteorology, geology, psychology, chemistry}



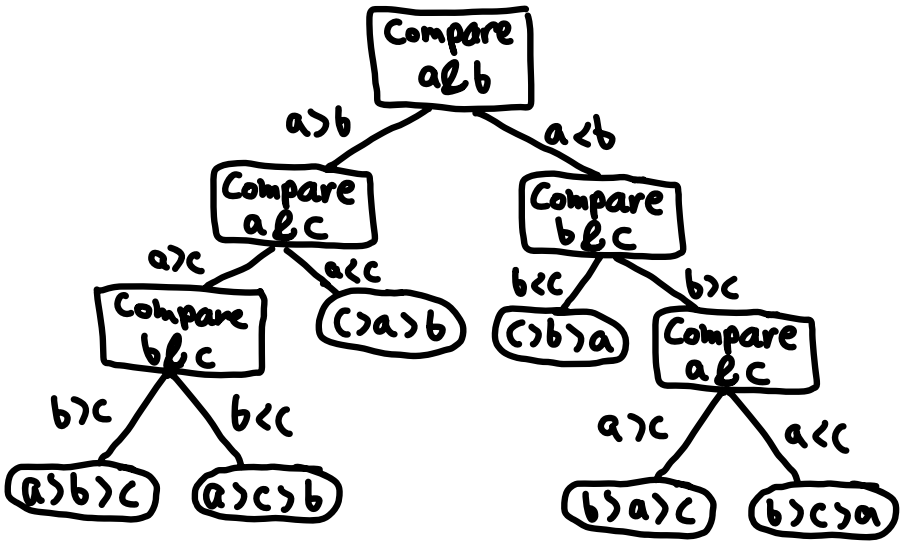
+ finish

We can use this tree to read off the words in order or to determine whether a word is in the list (e.g. geology, oceanography)

Decision trees

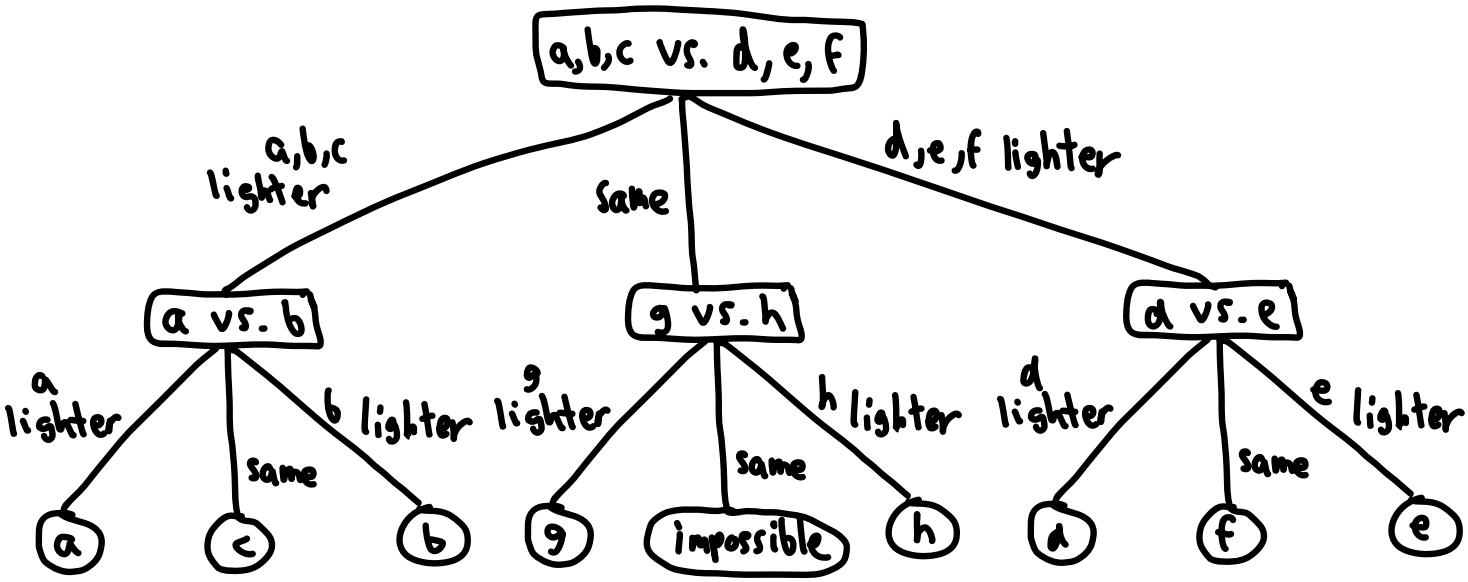
A decision tree is a rooted tree where each internal vertex corresponds to a decision, and this decision moves us to one of the two children. The leaves are the "conclusions"

Ex 4: a, b, c are three distinct numbers. Give a decision tree that orders a, b, c.



Ex 3: Suppose there are 7 coints, all w/ the same weight, and a counterfeit coin that weighs less than the others. How many weighings are needed to determine the counterfeit coin?

Coins: a, b, c, d, e, f, g, h



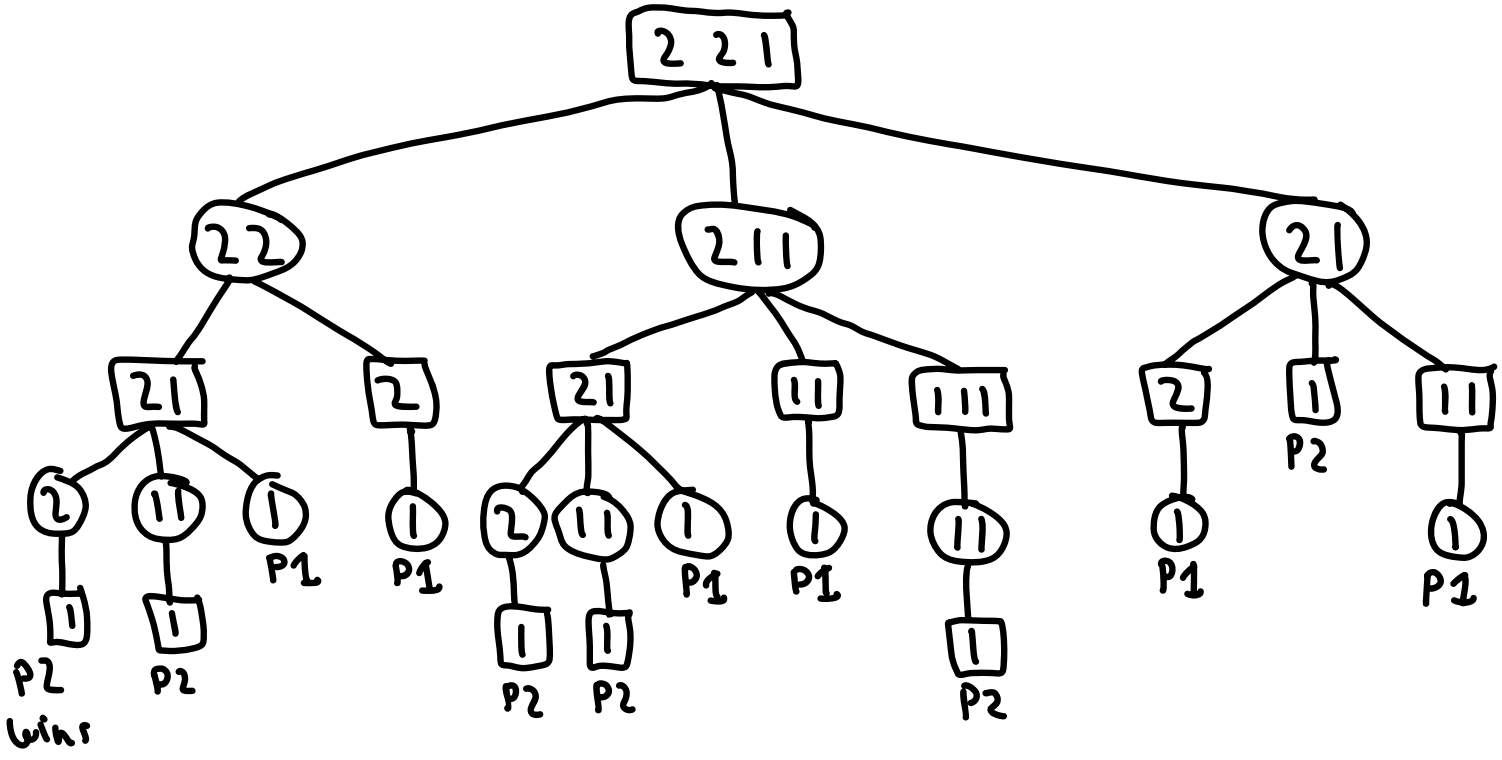
Game trees

A game tree is a decision tree where the decisions are made at the discretion of 2 or more (alternating) players

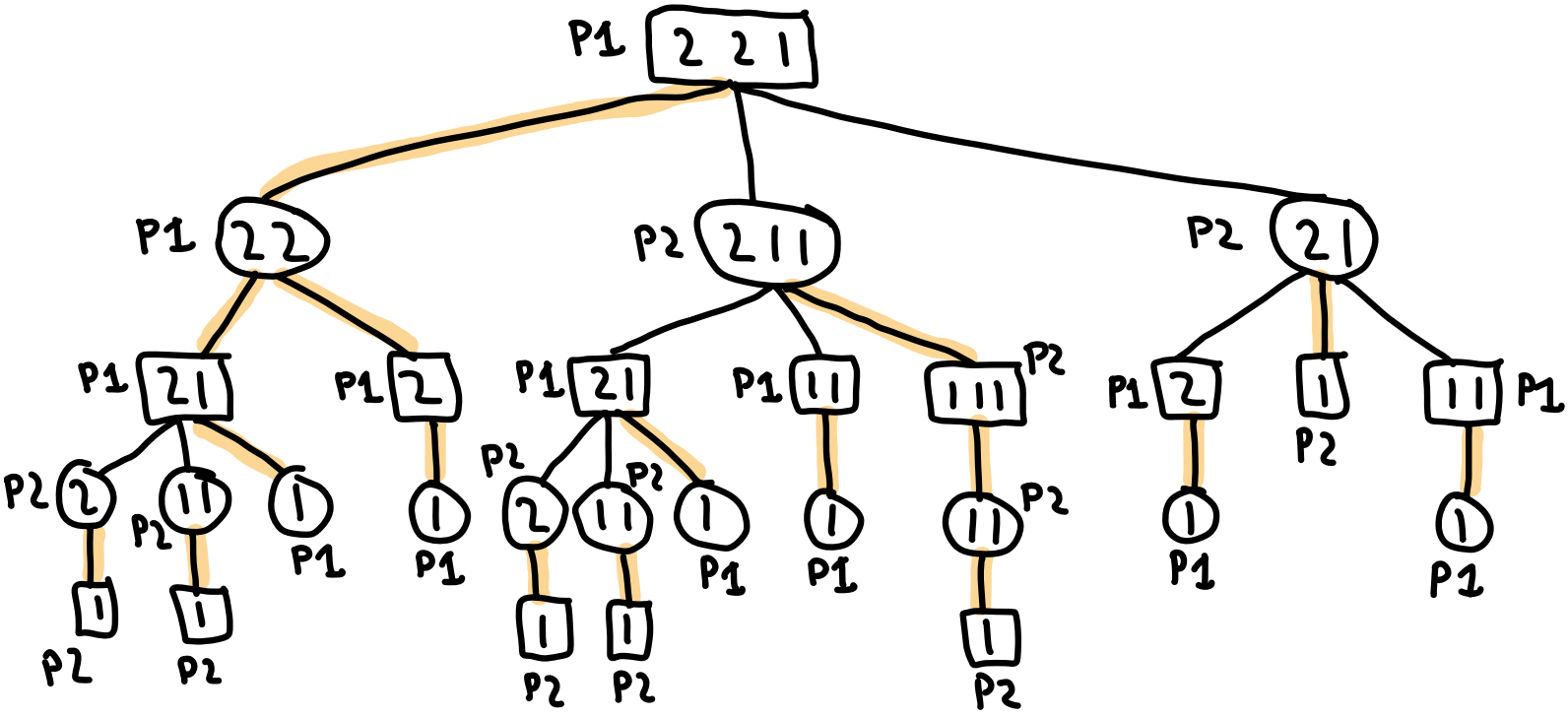
Ex 6: Nim:

- 2 players
- Several piles of stones
- Players alternate turns
- For each turn, a player takes 1 or more stones from one of the piles such that at least one stone remains
- The first player who can't move loses!

Start w/ three piles, containing 2, 2, 1 stone(s)



Can use the game tree to see who can force a win, and how



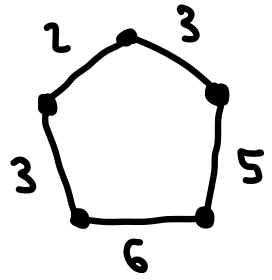
: best move(s)

§11.4/5: Spanning trees

Def: A spanning tree of a graph G is a subgraph which is a tree and contains all the vertices of G.

If G is a weighted graph, a spanning tree T is <u>minimal</u> if the sum of the edge weights of T is the smallest such value for any spanning tree of G (ties ok).

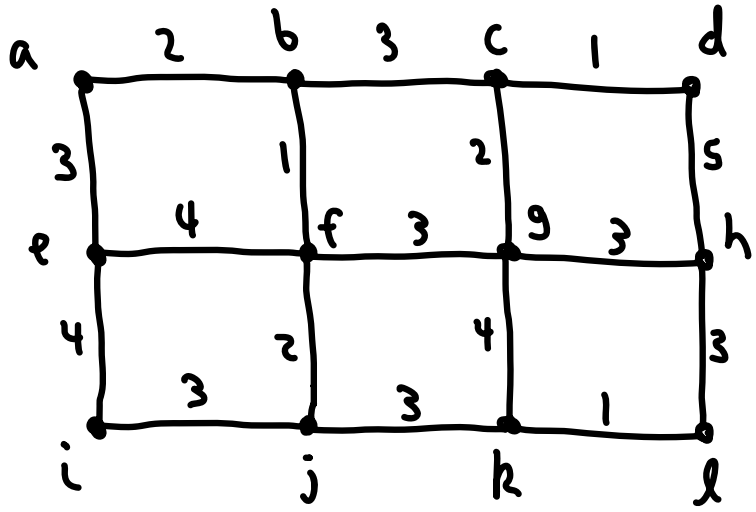
E.g. C_5



Spanning tree: remove any one edge

Minimal spanning tree:
remove the bottom edge

Class activity: Try to find a minimal spanning tree



We can build up a minimal spanning tree edge by edge:

Two algorithms:

Kruskal: Always add an edge w/ smallest wt. that doesn't create a simple circuit

Prim: Always add an edge w/ smallest wt. that is incident to the built-up part of the spanning tree and doesn't create a simple circuit

Both algorithms always give a minimal spanning tree